

8(a)	magnitude: (force =) Bqv	B1
	direction: $P \rightarrow Q$ or $E \rightarrow F$ or $S \rightarrow R$ or $H \rightarrow G$	B1
8(b)(i)	EHSP and FGRQ	B1
8(b)(ii)	PE or QF or RG or SH	B1
8(c)(i)	<i>any one correct starting point from:</i> • (mass of 1 atom =) $27 \times 1.66 \times 10^{-27}$ • (amount of substance per unit volume =) $2.7 / 27$ • 27 g (of substance) contains 6.02×10^{23} atoms • (2.7 g mass contains) 0.1 mol • (1 cm³ volume contains) 0.1 mol • (1 m³ volume contains) 10^5 mol	C1
	$n = (2.7 \times 10^3) / (27 \times 1.66 \times 10^{-27}) = 6.0 \times 10^{28}$ or $n = (2.7 / 27) \times 10^6 \times 6.02 \times 10^{23} = 6.0 \times 10^{28}$	A1
8(c)(ii)	$V_H = (0.15 \times 4.6) / (6.0 \times 10^{28} \times 0.090 \times 10^{-3} \times 1.60 \times 10^{-19})$	C1
	$= 8.0 \times 10^{-7} \text{ V}$	A1